P.S.R. ENGINEERING COLLEGE, SIVAKASI – 626 140
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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
QUESTION BANK

Programme:	B.E.	Branch	Electronics and Communication Engineering
Academic Year:	2024-2025	Year/Semester	II/IV
Course Code:	231EC45	Course Name	JAVA Programming
Course Tutor:	Mr.D.Arun Shunmugam, AP/ECE Ms.A.Madhumathi AP/ECE		

UNIT 1 - JAVA FUNDAMENTALS

PART - A

S.No	Questions	**Bloom's category Level**	**Course outcome**
1.	Who invented Java Programming? a) Guido van Rossum b)James Gosling c)Dennis Ritchie d)Bjarne Stroustrup **Answer: b**	K1	CO1
2.	Which one of the following is not a Java feature? a) Object Oriented b) Use of Pointers c)Portable d) Dynamic & Extensible **Answer: b**	K1	CO1
3.	The while loop repeats a set of code while the condition is not met? a)True b) False **Answer: b**	K1	CO1

4.	What will be the output of the following Java code?
	class increment {
	public static void main(Stringargs[])
	{
	int g = 3;
	System.out.print(++g * 8);
	}
	}
	a)23 b) 32 c)33 d)24 **Answer: b**
		K1	CO1

5.	Which statement is true about Java? a)Java is a sequence-dependent programming language b) Java is a code dependent programming language c) Java is a platform-dependent programming language d) Java is a platform-independent programming language **Answer: c**	K1	CO1
6.	What is the extension of java code files? a) .js b) .txt c) .class d) .java **Answer: d**	K1	CO1

7.	Which of the following is a type of polymorphism in Java Programming? a) Multiple polymorphism b) Compile time polymorphism c) Multilevel polymorphism d) Execution time polymorphism **Answer: b**	K1	CO1
8.	What will be the output of the following Java program? a) 1 2 3 4 5 b) 1 2 3 4 c) 1 2 d) 1 2 3 **Answer :d**	K1	CO1
9.	What is the purpose of control statements in Java? a) To perform mathematical calculations b) To create objects c) To declare variables d) To control the flow of execution in a program **Answer: d**	K1	CO1
10.	What is an Array in Java? a) A collection of elements with different types b) A collection of elements with the same type c) A resizable data structure d) A container for storing key-value pairs **Answer: b**	K1	CO1
11.	How do you access a element in an array in java/ a) By using the element's value b) By using the element's index c) By using the element's key d) By using the element's label **Answer: b**	K1	CO1
12.	Which of the following is used to perform bitwise XOR in java? a)% b) & c)^ d)” **Answer: c**	K1	CO1
13.	What is the result of the expression 2 + 2 * 2 / 2 – 2? a)2 b) 0 c)4 d)-2 **Answer: a**	K1	CO1
14.	Which component is used to compile, debug and execute java program? a) JVM b) JDK c) JIT d) JRE **Answer: d**	K1	CO1

15.	What is class in java? a) static reference b) template or blueprint c) instance of class d) None of above **Answer: b)**	K1	CO1
PART-B (2 marks)			
1.	List the features of Java Programming	K1	CO1
2.	Define classes and Objects	K1	CO1
3.	Define Polymorphism.	K1	CO1
4.	Compare Object Oriented Programming and Procedure Oriented Programming.	K2	CO1
5.	What are the datatypes available in java ?	K1	CO1
6.	How will you declare and access the elements of an array?	K2	CO1
7.	What is meant by Static binding and Dynamic binding?	K1	CO1
8.	Write a simple java code to add two numbers.	K2	CO1
9.	Write a java code to check the number is positive or negative.	K2	CO1
10.	Show the code to check the given number is od or even.	K2	CO1
11.	Define variable and how will you declare ?	K2	CO1
PART-C (14 marks)			
1.	Outline the working of Java Virtual Machine (JVM) Architecture.	K2	CO1
2.	Explain about different data types in java with examples for each.	K2	CO1
3.	Explain about the different types of Operators in java with examples.	K2	CO1
4.	Summarize the OOPS Concepts.	K2	CO1
5.	Demonstrate java program for the following: (i) to print Fibonacci series within given range. (ii)to print the Factorial of a given number using do while loop	K2	CO1
6.	Infer Java programs using Control structures with examples for each.	K2	CO1

7.	Demonstrate java program for the following: (i)to print prime numbers between the given range. (ii)to compare two numbers	K2	CO1

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UNIT –II JAVA CLASSES

PART - A			
1.	Which of these keywords are used to define an abstract class? a)Abstract b)abstract c)abst d)static Answer: b	K1	CO2
2.	In JAVA, can a class implement multiple interfaces? Interfaces cannot be implemented by classes Only abstract classes can implement interfaces No, a class can implement only one interface Yes, a class can implement multiple interfaces Answer: d	K1	CO2
3.	Which of these packages contains abstract keyword? a) java.lang b) java.util c) java.io d) java.system Answer: a	K1	CO2
4.	Arrays in Java are implemented as? a) class b) object c) variable d) none of the mentioned Answer: b	K1	CO2
5.	_____ keyword is used to prevent content of a variable from being modified? a) final b) last c) constant d) static Answer: a	K1	CO2
6.	_____ cannot be declared static? a) class b) object c) variable d) method Answer: b	K1	CO2
7.	Which of these access specifiers can be used for a class so that its members can be accessed by a different class in the different package? a) Public b) Protected c) Private d) No Modifier Answer: a	K1	CO2

8.	What will be the output of the following Java program?
	package pkg;
	class output
	{
	public static void main(String args[])
	{
	StringBuffer s1 = new StringBuffer("Hello");
	s1.setCharAt(1, x);
	System.out.println(s1);
	}
	}
	a) xello b) xxxxx c) Hxllo d) Hexlo Answer: c
		K1	CO2

9.	When does Exceptions in Java arises in code sequence? a) Run Time b) Compilation Time c) Can Occur Any Time d) None of the mentioned Answer: a	K1	CO2
10.	Which of these keywords is used to manually throw an exception? a) try b) finally c) thrown d) catch Answer: c	K1	CO2
11.	What is an exception in Java? a) An error that occurs during compilation b) An unexpected event that occurs during program execution c) A logical error in the code d) A warning issued by the compiler Answer: b	K1	CO2
12.	Which class is the superclass of all exceptions in Java? a) Error b) Runtime Exception c) Throwable d) Exception Answer: c	K1	CO2
13.	Which block must follow a try block? a) final b) catch c) done d) except Answer: b	K1	CO2
14.	Which of the following is used to group multiple exceptions in a single catch block? a) Using the | operator b) Using the & operator c) Using the && operator d) Using the || operator Answer: a	K1	CO2
15.	What is the use of try & catch? a) It allows us to manually handle the exception b) It allows to fix errors c) It prevents automatic terminating of the program in cases when an exception occurs d) All of the mentioned Answer: d	K1	CO2
16.	Which of these keywords are used for the block to be examined for exceptions? a) try b) catch c) throw d) check Answer: a	K1	CO2
17.	What will be the output of the following program? int x=5, y=10; System.out.println(x+y);	K1	CO2

	a) 5 b) 10 c) true d) None of the above Answer: d		
18.	Which of the following is valid syntax to declare an array in Java? a) int arr[] b) arr int[] c) array int d) int[] arr; Answer: a	K1	CO2
19.	Which operator is used for concatenation of Strings in Java? a) + b) & c) && d) % Answer: a	K1	CO2
20.	Which data type is used to store a single character in Java? a) char b) String c) byte d) Boolean Answer: a	K1	CO2

PART-B (2 marks)			
1.	What are different types of access modifiers (Access specifiers)?	K1	CO2
2.	How to define and import a package in Java.		
3.	Define Abstract class	K1	CO2
4.	Compare static and inner class.	K2	CO2
5.	Explain the usage of Java packages.	K2	CO2
6.	What is a package? How to define a package?	K1	CO2
7.	What is the difference between this () and super ()?	K2	CO2
8.	List the Wrapper class in Java.	K1	CO2
9.	What is Exception Handling in Java?	K2	CO2
10.	What is the use of try and catch block in java?	K2	CO2
11.	What are runtime exceptions?	K1	CO2
12.	Demonstrate how to declare a Book class with title and author attributes, create an object, and print the values.	K2	CO2
13.	Demonstrate using throw and throws in a simple method.	K2	CO2
14.	Demonstrate how to define and import a package.	K2	CO2
15.	Explain how to create a Java package named library with classes Book and Member	K2	CO2
16.	Develop a Java program that uses an interface to implement multiple behaviours for a Vehicle class.	K3	CO2
17.	Show how to import the Book class from the library package in another class.	K2	CO2
PART-C (14 marks)			
1.	Explain the concept of abstract classes in Java, describe their properties, and write a program where an abstract class Shape is extended by Circle and Rectangle classes. Display the area of each shape.	K2	CO2
2.	Build a Java Program to implement method Overloading and method overriding	K3	CO2
3.	Explain inner classes in Java and their importance. Write a program with a class Student having an inner class Address. Display student details along with the address.	K2	CO2
4.	Explain **interfaces** in Java and how they are implemented and Create an interface Library with methods drawbook (), return Book(), check Status(), and reserve Book(), and implement it in a class City Library.	K2	CO2

5.	Demonstrate the exception handling mechanism in Java using a program with try, catch, and finally blocks to handle Arithmetic Exception and ArrayIndexOutOfBoundsException.	K2	CO2
6.	How to handle multiple catch blocks for a nested try block with a suitable example program.	K1	CO2
7.	Summarize the uses of throw and throws clauses in Java exception handling and give suitable example program.	K2	CO2
8.	Develop a Library interface which has draw book (), return book () (with fine), check status () and reserve book () methods. All the methods tagged with public.	K3	CO2
9.	Develop a static inner class called Pair which has a minMax method for finding min and max values from an array.	K3	CO2
10.	Explain the Java exception hierarchy with a diagram and give examples of checked and unchecked exceptions.	K2	CO2

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UNIT- III I/O PACKAGE

PART – A			
1.	Which of this class is used to read characters in a file? a) FileReader b) File Writer c) FileInputStream d) InputStreamReader Answer: a	K1	CO3
2.	Which of this stream contains the classes which can work on character stream? a) InputStream b) OutputStream c) Character Stream d) All of the mentioned Answer:c	K1	CO3
3.	What type of data can be read by an InputStream? a) Characters b) Raw bytes c) Strings d) Objects Answer:b	K1	CO3
4.	_____is a process of extracting/removing the state of an object from a stream. a) Serialization b) Externalization c) File Filtering d) Deserialization Answer:d	K1	CO3
5.	Which of this interface extends DataInput interface? a) Serializable b) Externalization c) ObjectOutput d) ObjectInput Answer:d	K1	CO3
6.	State whether the following statements about the stream in Java. i) The two basic streams used are the input and output streams. ii) Filters are used to read data from one stream and write it to another stream. A) True, True B) True, False C) False, True D) False, False Answer:a	K1	CO3
7.	_____ streams provide functionally for threads to communicate and exchange data between them. A) Object B) Piped C) Pushback D) Filtered Answer:b	K1	CO3

8.	Multithreaded programming is a process in which _____ a) two different processes run simultaneously b) two or more parts of same process run simultaneously c) many different process are able to access same information d) single process can access information from many source Answer:b	K1	CO3
9.	Thread priority in Java is _____ a) Integer b) Float c) double d) long Answer:a	K1	CO3

10.	What is the name of the thread in the following Java Program?
	
	`class multithreaded_programing`
	`{`
	`public static void main(String args[])`
	`{`
	`Thread t = Thread.currentThread();`
	`System.out.println(t);`
	`}`
	`}`
	
	a) main b) Thread c) System d) None of the mentioned
	
	Answer: b
		K1	CO3

11.	Which method is used to start a thread in Java? a) `run()` b) `start()` c) `execute()` d) `init()` Answer: b	K1	CO3
12.	Which of the following is a disadvantage of multi-threading? a) Improved performance b) Reduced CPU utilization c) Complexity in debugging and testing d) Simpler code structure Answer:c	K1	CO3
13.	What is the primary advantage of multi-threading in Java? a) Better resource management b) Simultaneous execution of tasks c) Enhanced debugging capabilities Answer: b d) Decreased memory usage	K1	CO3
14.	Which of the following best describes the life cycle of a thread in Java**?** a) New -> Ready -> Running -> Waiting -> Terminated b) New -> Blocked -> Runnable -> Terminated c) Ready -> Running -> Waiting -> Blocked -> Terminated d) New -> Waiting -> Terminated -> Blocked Answer: a	K1	CO3
15.	What is the potential issue with thread synchronization in multi-threading?	K1	CO3

	a) Deadlock b) Parallelism c) Thread starvation d) Faster execution Answer: a		
16.	In multithreading, which method allows a thread to give up CPU and let other threads run? a) sleep() b) yield() c) wait() d) join() Answer: b	K1	CO3
17.	Which class is used to read primitive data types from a stream? a) ObjectInputStream b) DataInputStream c) FileReader d) BufferedReader Answer: b	K1	CO3
18.	Which of the following is true about **FileWriter** in Java? a) Reads character streams b) Writes character streams c) Writes byte streams d) Reads byte streams Answer: b	K1	CO3
19.	Which exception is thrown when a thread is waiting, sleeping, or otherwise paused for a long time and another thread interrupts it? a) InterruptedException b) IOException c) RuntimeException d) FileNotFoundException Answer: a	K1	CO3
20.	Which method is used to pause the execution of a thread for a specified time? a) wait() b) sleep() c) yield() d) stop() Answer: b	K1	CO3

PART-B (2 marks)			
SL.NO	**QUESTION**	**BTL**	**CO**
1.	What is a stream? Explain its types.	K1	CO3
2.	List classes that are used by Byte streams for input and output operation.	K1	CO3
3.	List classes that are used by character streams for input and output operation	K1	CO3
4.	What is race condition?	K1	CO3
5.	Define Object Serialization.	K1	CO3
6.	Define Serialization and Deserialization in java with example.	K2	CO3
7.	What is multithreading? State its advantages.	K1	CO3
8.	What is thread?	K1	CO3
9.	Compare thread,process and task.	K2	CO3
10.	Outline thread lifecycle.	K1	CO3
11.	What is the difference between InputStream and OutputStream in Java?	K1	CO3
12.	Explain the concept of object serialization in Java.	K2	CO3
13.	What is the use of ObjectOutputStream and ObjectInputStream?	K1	CO3
14.	Define deserialization. How is it different from serialization?	K1	CO3
15.	What is the purpose of a FileReader class in Java?	K2	CO3
16.	How can you copy a file using Java I/O classes?	K1	CO3
17.	Explain the use of Filter streams in Java I/O.	K2	CO3
18.	What are Pipe Streams in Java?	K1	CO3
19.	What is the function of the BufferedReader class in Java I/O?	K2	CO3
20.	How can you read data from a file using a Scanner class in Java?	K1	CO3

SL.NO	QUESTION	BTL	CO
1.	Explain the concepts of Input Streams and Output Streams in Java with suitable examples. Also, discuss object serialization and deserialization, providing a sample program demonstrating object serialization and deserialization to a file.	K3	CO3
2.	Describe the life cycle of a thread in Java. Discuss the advantages and common issues associated with multi-threading. Write a simple Java program to create and run a thread, and explain how thread synchronization can be implemented to avoid thread interference.	K3	CO3
3.	Describe the **life cycle of a thread** in Java. Explain the different states a thread can be in, and the transitions between these states with an example or diagram.	K2	CO3
4.	Explain the concept of **Input Streams** and **Output Streams** in Java. Discuss the various classes used for input and output operations in Java and provide a sample program demonstrating how to read and write data from/to files.	K2	CO3
5.	Explain the concept of **Filter Streams** and **Pipe Streams** in Java. Provide an example of how they are used for filtering or piping data between threads.	K3	CO3
6.	Discuss the **advantages** and **issues** associated with using multi-threading in Java. Explain how multi-threading can improve performance and the potential problems that may arise, such as thread interference.	K2	CO3
7.	Develop a Swing program with a JFrame containing a JButton and a JTextField. When the button is clicked, the text in the text field should be displayed in a JLabel.	K3	CO3
8.	Develop a Java program demonstrating the use of FlowLayout and GridLayout. Place multiple buttons in a frame and show how the layout manager affects their placement.	K3	CO3
9.	Construct a Java applet that reads a numeric parameter from HTML and displays its square on the applet window. Build a simple AWT applet that draws a rectangle and a circle on the screen.	K3	CO3

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UNIT IV: GRAPHICAL USER INTERFACE (GUI)

PART-A			
S.NO	**QUESTION**	**BTL**	**CO**
	Which of the following is a feature of the AWT (Abstract Window Toolkit) in Java? A) Lightweight components B) Heavyweight components C) No event handling D) Cannot be used for GUI development Answer: b	K1	CO4
	Which of the following layout managers arranges components in a grid of rows and columns? A) FlowLayout B) BorderLayout C) GridLayout Answer: c	K1	CO4
	Which method is called when an applet is first loaded into memory? A) start() B) init() C) stop() D) destroy() Answer: b	K1	CO4
	Which of the following is true about Swing components? A) Swing components are heavyweight. B) Swing components are dependent on the underlying operating system. C) Swing components are written entirely in Java. D) Swing components are only used for event handling. Answer: c	K1	CO4
	What is the purpose of the ActionListener interface in Java? A) To handle mouse events. B) To listen for key events. C) To handle button clicks and other action events. D) To manage window resize events. Answer: c	K1	CO4
	In an applet, how do you pass parameters through an HTML tag? A) Using <param> tag within <applet> tag. B) Using the setParameter method. C) Parameters are passed automatically. D) By defining variables within the applet's Java code Answer: a	K1	CO4
	Which method of the JFrame class is used to set the window's title? A)setTitle() B) setWindowTitle() C) windowTitle() D) title() Answer: a	K1	CO4
	Which of the following methods in an applet is responsible for cleaning up resources when the applet is destroyed? A) start() B) init() C) stop() D) destroy() Answer: d	K1	CO4

	Which of the following components is a container in Swing? A) JLabel B) JButton C) JTextField D) JPanel Answer: d	K1	CO4
	Which of the following Swing components is used to create a button? A) JButton B) JTextField C) JList D) JCheckBox Answer: a	K1	CO4
	Which layout manager arranges components in five regions: North, South, East, West, and Center? A) GridLayout B) BorderLayout C) FlowLayout D) CardLayout Answer: b	K1	CO4
	Which method is used to make a JFrame visible? A) show() B) visible() C) setVisible(true) D) display() Answer: c	K1	CO4
	What is the default layout manager for a JPanel in Swing? A) BorderLayout B) FlowLayout C) GridLayout D) CardLayout Answer: b	K1	CO4
	Which of the following is not a method of the ActionListener interface? A) actionPerformed() B) actionOccurred() C) addActionListener() D) removeActionListener() Answer: b	K1	CO4
	Which component is used to create a drop-down list in Swing? A) JList B) JComboBox C) JRadioButton D) JCheckBox Answer: b	K1	CO4
	Which of the following is a valid layout manager in AWT? A) FlowLayout B) GridBagLayout C) BorderLayout D) All of the above Answer: d	K1	CO4
	What is the purpose of the ActionListener interface in Java? A) It handles keyboard events. B) It handles mouse events. C) It handles action events such as button clicks. D) It handles window events Answer: c	K1	CO4
	Which method is automatically called when an applet is started? A) init() B) start() C) stop() D) destroy() Answer: b	K1	CO4
	Which of the following is a container class in Swing? A) JButton B) JTextField C) JPanel D) JLabel Answer: c	K1	CO4
	Which method in AWT is used to handle button clicks? A) buttonPressed() B) actionPerformed() C) handleClick() D) actionEvent() Answer: b	K1	CO4

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PART B(2-Mark Questions)			
S.NO	**QUESTION**	**CO**	**BTL**
1.	What is AWT in Java, and how does it differ from Swing?	CO4	K1
2.	List the basic components provided by AWT for GUI development.	CO4	K1
3.	Explain the difference between lightweight and heavyweight components in Java.	CO4	K1
4.	Describe the role of a layout manager in AWT and Swing.	CO4	K1
5.	Write a simple Java program using AWT to create a window with a button.	CO4	K1

6.	How do you handle button click events using ActionListener in AWT?	CO4	K1
7.	Compare and contrast the FlowLayout and BorderLayout layout managers in AWT.	CO4	K2
8.	Analyze the life-cycle of an applet and explain each of its stages.	CO4	K2
9.	Evaluate the performance differences between AWT and Swing components for GUI applications in Java.	CO4	K2
10.	Assess the role of JApplet in modern web development and its alternatives.	CO4	K2
11.	What is the purpose of the init() method in the applet life-cycle?	CO4	K1
12.	What is the difference between the JButton and JCheckBox in Swing?	CO4	K1
13.	Explain how the JComboBox component is used to create a drop-down list in Swing.	CO4	K2
14.	How do you pass parameters from an HTML page to an applet using the <param> tag?	CO4	K1
15.	Write an AWT program to demonstrate the use of multiple layout managers in a single window.	CO4	K1
16.	Demonstrate how you can embed an applet in an HTML page with parameters.	CO4	K2
17.	Compare between JPanel and JFrame in terms of functionality in a Swing-based application.	CO4	K2
18.	How would you implement event handling for a JTextField to perform an action when the user presses the "Enter" key?	CO4	K1
19.	Create a Java Swing application that uses a JFrame to display a JButton and JLabel. When the button is clicked, the label’s text should change.	CO4	K2
20.	Design a Swing form using JTextField, JButton, and JLabel to accept a user's name and display it upon clicking a button.	CO4	K2

PART C 14 Mark Questions			
S.No	**QUESTION**	**CO**	**BTL**
1.	Explain the differences between **AWT** and **Swing** in Java.(7) **b)** Describe the types of components available in **AWT** and give examples of when each might be used.(7)	CO4	K3
2.	**a)** Develop a Java program using **AWT** to create a simple user interface that contains a **text field** and a **button**. When the button is clicked, the text entered in the text field should be displayed in a **label** below the text field.(7) **b)** Analyze the event handling process in the above example. How is the button click event captured and processed?(7)	CO4	K2
3.	**a)** Design a Java program that uses **Swing components** to create a login form with two JTextFields (for username and password) and a JButton (for login). When the button is clicked, the entered username and password should be displayed in a dialog box.(7) **b)** Explain how the **Swing layout manager** is used to position the components within the JFrame.(7)	CO4	K3
4.	**a)** Describe the **Applet life-cycle** with respect to the following methods: init(), start(), stop(), and destroy().(7) **b)** Write a small program using **Applet** that demonstrates the use of these methods and shows how parameters can be passed from an HTML page to the applet.(7)	CO4	K3
5.	**a)** Discuss the advantages and disadvantages of using **AWT** vs **Swing** for building modern Java GUI applications.(7) **b)** Design a GUI using **Swing** to demonstrate the use of **JComboBox**, **JRadioButton**, and **JCheckBox**. The GUI should allow the user to select a color from the JComboBox and use checkboxes to select options. Display the selections when the button is clicked.(7)	CO4	K3
6.	Explain the concept of **layout managers** in Java. Compare the **FlowLayout**, **BorderLayout**, and **GridLayout** managers in terms of their use cases and how they differ in arranging components.(7) **b)** Create a user interface using **Swing** that demonstrates the use of **BorderLayout** and **GridLayout** together to arrange buttons and text fields in a panel.(7)	CO4	K2
7.	**a)** Discuss the **Swing** component JFrame in terms of creating a window-based application. What are its key features, and how does it interact with other Swing components?(7) **b)** Create a simple **Swing-based application** that uses **JFrame** to display a title, a text input field, and a button. When the button is clicked, the text entered in the field should be displayed in a message dialog.(7)	CO4	K3